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THERMODYNAMICS 1(MEC 122)

FUNDAMENTAL CONCEPTS AND LAWS OF THERMODYNAMICS

Introduction

Thermodynamics is a branch of mechanical engineering that is concerned with the relationship between heat, work and those properties of matter that relate to heat and work. Note: One of the main goals of these notes is to formalize with the relationship between heat, work, and energy.

The name *thermodynamics* comprises two different words: *thermo* (heat) and *dynamics* (power/transfer of heat/energy). Therefore in other words, thermodynamics is the science of understanding or studying the relationship between heat, work, energy, and temperature of engineering systems.

In term of area of application, thermodynamics is commonly encountered everywhere in life such as the pumping blood in your body. In **household applications** includes: cooking stove, pressure cooker, Air conditioning, iron, TV, Mobile, heater refrigerators, etc. **Industrial application** are: automotive vehicles airplanes jet engines, nuclear power plants, Alternative energy (solar, wind, Biomass), cooling, heating systems etc. Thermodynamics, basically entails *four laws* known as Zeroth, First, Second and third law of thermodynamics.

Whenever there is an interaction between energy and matter, thermodynamics is involved.

System, Boundary, Working fluid, and surroundings of a System

A **system** is defined as *a quantity of matter or a region in space chosen for study*. The mass or region outside the system is called the **surroundings**. The real (boundary of a water tank) or imaginary (boundary of a nozzle, opened from both ends) surface that separates the system from its surroundings is called the **boundary** (Figure.1). The boundary of a system can be *fixed* (e.g *potato*) or *movable* (eg *piston cylinder*). Note that the boundary is the contact surface shared by both the system and the surroundings. Mathematically speaking, the boundary has zero thickness, and thus it can neither contain any mass nor occupy any volume in space. Thus, a thermodynamic system: a quantity of fixed mass under investigation



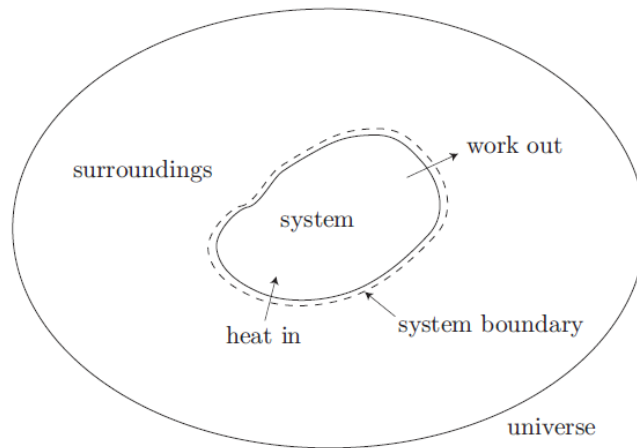


Figure.1 System, surroundings, boundary

A universe composed of a system, its surroundings, and the system boundary. We allow two important interactions between the system and its surroundings:

- heat can cross into the system (our potato can get hot), and
- Work can cross out of the system (our potato can expand).

Now, the system boundaries can change, for example the potato might expand on heating, but we can still distinguish the system and the surroundings.

Thermodynamics Working fluid

Behaviour of the working substance is very essential factor in understanding thermodynamics. In this course, focus is given to pure substances such as gases and steam properties and how they are interrelated are important in the design and operation of thermal systems. Heat engines and other cyclic devices usually involve a fluid to and from which heat is transferred while undergoing a cycle. This fluid is called the **working fluid**. The working fluid can be either a gas or a liquid. The working fluid used in the refrigeration cycle is called a refrigerant. A working fluid can be a pressurized gas or liquid that actuates a machine. Examples include steam in a steam engine, air in a hot air engine and hydraulic fluid in a hydraulic motor or hydraulic cylinder.

More generally, in a thermodynamic system, the working fluid is a liquid or gas that absorbs or transmits energy. Simply we describe working fluid as the **matter contained within boundaries of a system**. Matter can be in solid, liquid, vapor or gaseous phase. The working fluid in applied thermodynamic problems is either approximated by a perfect gas or a substance which exists as liquid and vapor. The state of the working fluid is defined by certain characteristics known as properties

Types of the thermodynamic systems

After defining the system and boundary it is so important to observe the interaction between the system and surroundings which takes place across the boundary. This interaction can be in form of mass as well as in form of energy. Based on the interaction across the boundary the system can be categorized into three types: closed, isolated, and opened systems (Figure.2), depending on whether a fixed mass or a fixed volume in space is chosen for study

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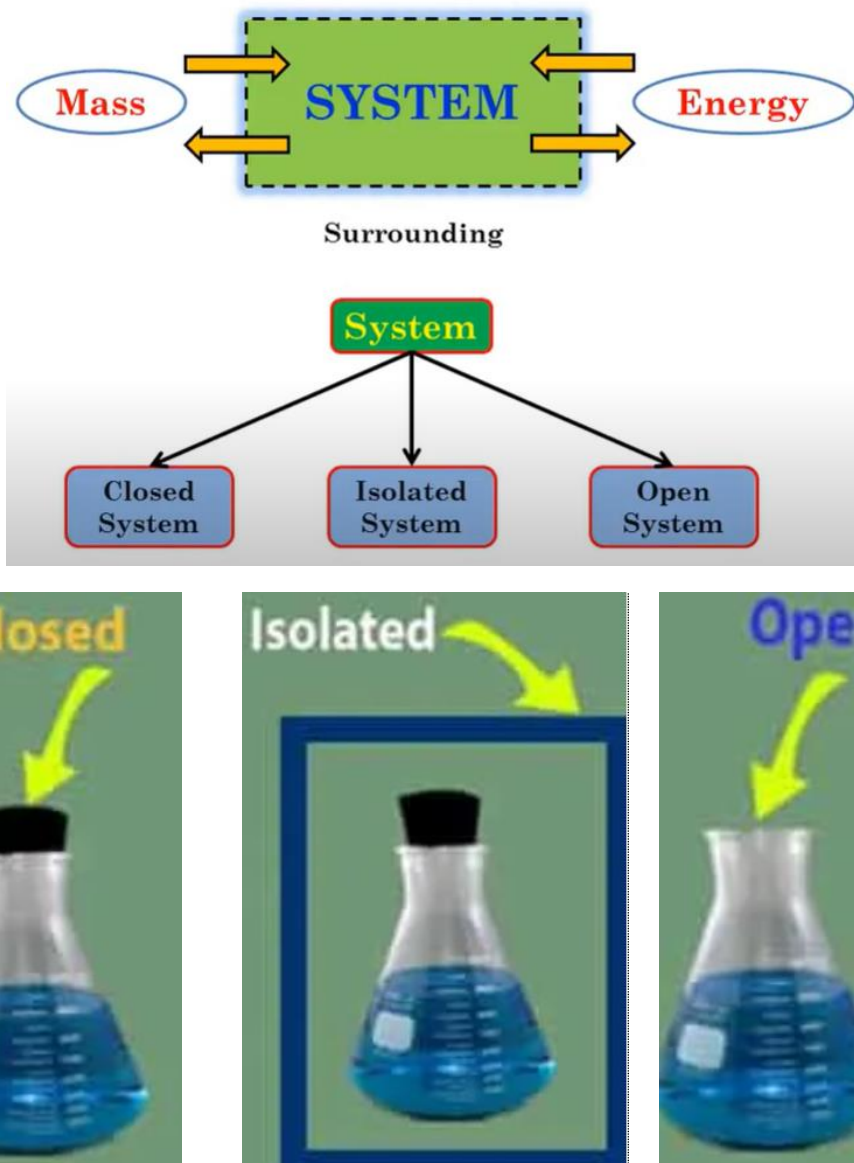


Figure.2 Types of the thermodynamic system in detail

Closed system

A closed system is a type of system which has only energy interaction with the surrounding but no mass interaction. For example, consider a tightly closed cylinder device (Figure 3). In this system, no mass can transfer in or out of the boundary but energy can transfer across the system boundary in the form of Heat or Work. It is also known as **control mass** because it consists of a fixed amount of mass, and no mass can cross its boundary. Note that the volume of a closed system does not have to be fixed.

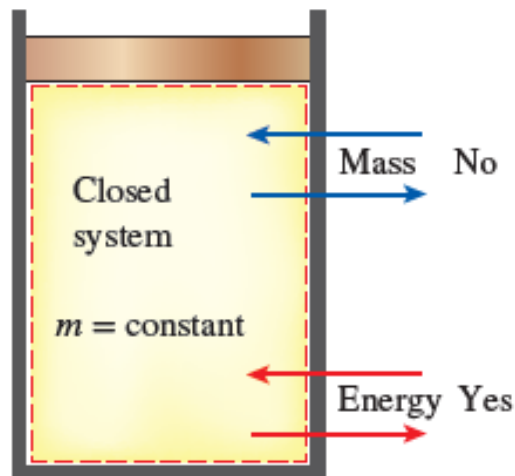


Figure 3 A closed system Mass cannot cross the boundaries of a closed system, but energy can

Example of a closed system with a moving boundary consider a tightly piston cylinder device (without valves) as shown in Figure 4. Let say we would like to find out what happen to the enclosed gas when it is heated. Gas inside the cylinder is our system. The inner surface of the piston cylinder forms the boundary. It is interesting to observe that this boundary is a moving boundary and the real boundary and since no mass is crossing the boundary this is called a closed system. Note that here energy is crossing the boundary in form of heat and the part of boundary may move because of the expansion of the gas. Everything outside the gas including the piston cylinder is call the surrounding.

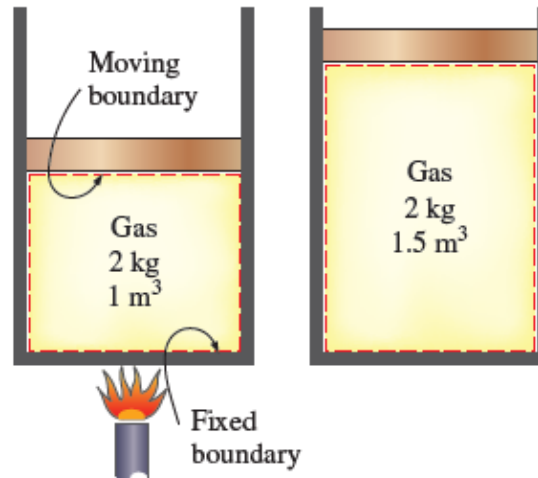


Figure.4 A closed system with a moving boundary

Keep some amount of fluid in the cylinder and close all the valves. By applying heat to the bottom of the cylinder it will start to expand the fluid (Figure 4) and as a result, the temperature of the fluid will gradually increase, and that heat will be emitted outside. So, we added energy into the system and energy transferred too. Hence, there is energy transfer although there was no mass transfer. Other examples include Water in a closed pressure vessel, cooking in a pressure cooker, Car Battery, Piston cylinder, a can of cold drinks

Isolated system

Isolation means totally separated from something. An isolated system is a system that not influenced by its surroundings so that there will not be any mass or energy transfer between the system and surrounding (Figure.5). Thermos Flask is an example of an isolated system.

We all have flasks in our homes. What do we do with it? We used to store hot water or hot tea or hot milk for a long time! How is it possible?

- Because this flask is well insulated, and it helps to prevent heat energy to go out and we get hot water.
- But have you observed the temperature of hot water after a long time is reduced and later on it will be same as normal temperature? Why?
- Because the insulation is not perfect. If the insulation is perfect, that means it will not allow any heat transfer across the boundary, then there will not be any energy transfer, any mass as well as energy transfer. thus, **an isolated system is a special case of a closed system**

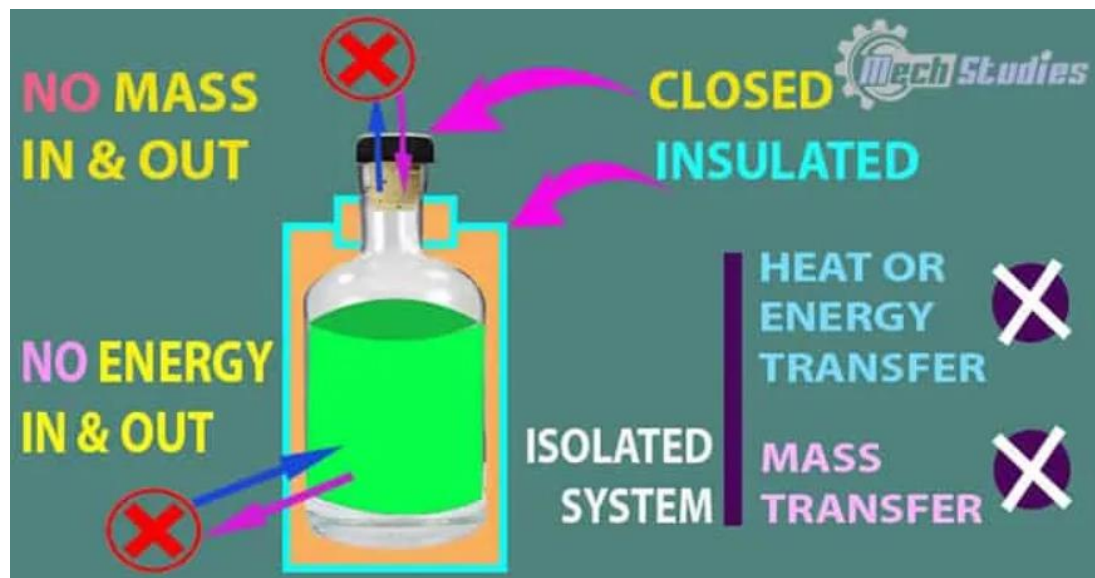


Figure.5 An Isolated system thermos Flasks

Open system

An open system or control volume, is a properly selected region in space. The system is called open because mass may flow in or out of the system such as a compressor, turbine, or nozzle. Flow through these devices is best studied by selecting the region within the device as the control volume. Both mass and energy can cross the boundary of a control volume. A large number of engineering problems involve mass flow in and out of a system and, therefore, are known as **control volumes (open systems)**.

Examples of open systems include Water in a pipe, a water heater, Engine oil in the engine, a car radiator Water Pump, the liquid in the heat exchanger, Boiler, Turbine, and compressor.

There are no concrete rules for the selection of control volumes, but the proper choice certainly makes the analysis much easier. For example, if we were to analyze the flow of air through a nozzle, a good choice for the control volume would be the region within the nozzle.

The boundaries of a control volume are called a **control surface**, and they can be real or imaginary. In the case of a nozzle, the inner surface of the nozzle forms the **real** part of the boundary, and the entrance and exit areas form the **imaginary** part since there are no physical surfaces there (Figure 6)

A control volume can be fixed in size and shape, as in the case of a nozzle, or it may involve a moving boundary, as shown in Figure 7. Most control volumes, however, have fixed boundaries and thus do not involve any moving boundaries. A control volume can also involve heat and work interactions just as a closed system, in addition to mass interaction

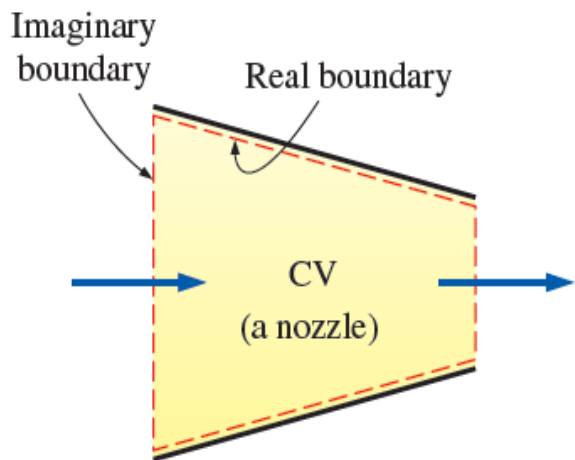


Figure 6 A control volume (CV) with real and imaginary boundaries

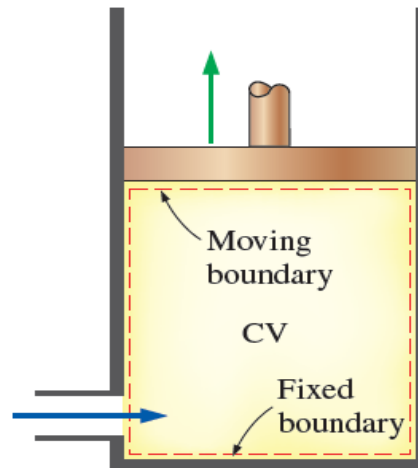


Figure 7 A control volume with fixed & moving boundaries as well as real and imaginary boundaries

A control volume can involve fixed, moving, real, and imaginary boundaries.

As an example of an open system, consider the water heater shown in Figure 8. Let us say that we would like to determine how much heat we must transfer to the water in the tank to supply a steady stream of hot water. Since hot water will leave the tank and be replaced by cold water, it is not convenient to choose a fixed mass as our system for the analysis. Instead, we can concentrate our attention on the volume formed by the interior surfaces of the tank and consider the hot and cold

water streams as mass leaving and entering the control volume. The interior surfaces of the tank are from the control surface for this case, and mass is crossing the control surface at two locations.

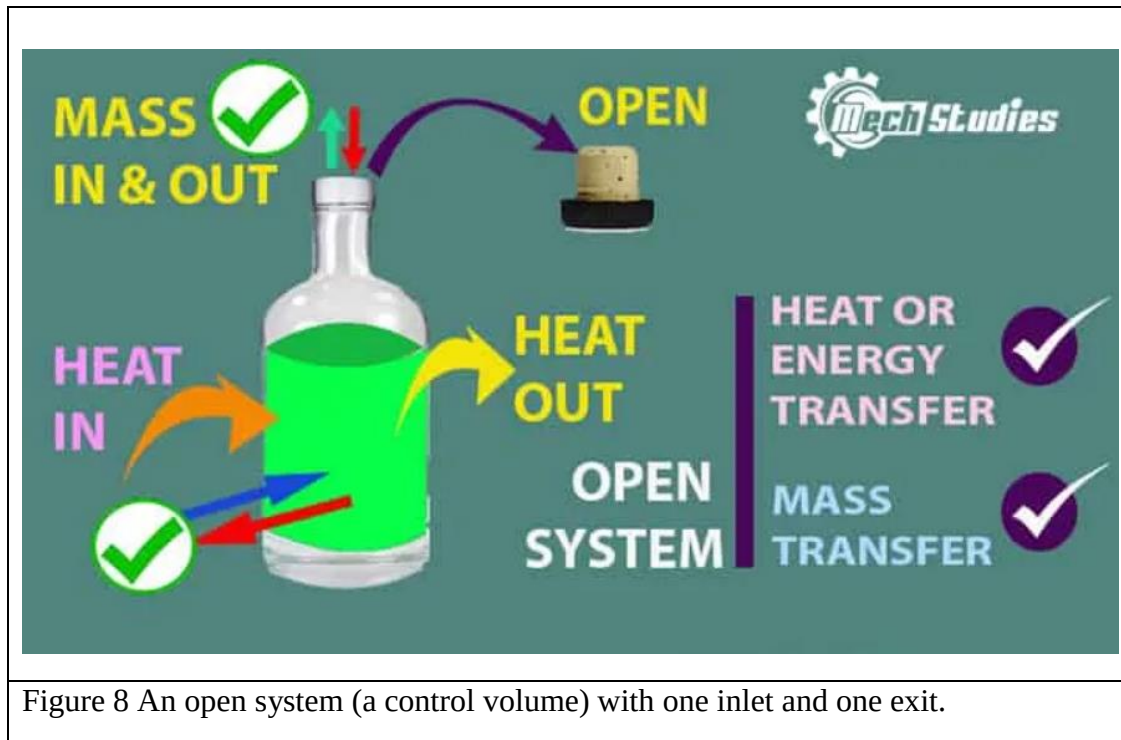


Figure 8 An open system (a control volume) with one inlet and one exit.

Summary:

Main differences between open, closed system and Isolated system				
S/N	Description	Open system	Closed system	Isolated system
1	Definition	It is the system, where mass or energy can be transferred	It is the system, where energy transfer happens but no mass transfer	It is the system, where no mass & no energy transfer
2	Energy transfer	yes	yes	No
3	Mass transfer	yes	No	No
4	Change in mass is possible or not	changes	No change	No change
5	boundary	Open	closed	closed

6	reality	It exists	It exists	It is theoretical
7	example	Making tea in an open container	keeping tea in a closed uninsulated container	keeping tea in a insulated thermo flask

Thermodynamic state, process and cycle

A **state** refers to the condition of a system as described by its properties. Any operation in which one or more of the properties of a system change is called a **change of state**. The succession of states passed through during the change of state is called the **path of the change of state**. When the path is completely specified, the change of state is called a **process**. Thus, a process is a transformation from one state to another (Figure 9), e.g constant pressure process. When all the properties of a system have definite values, the system is said to exist at a **definite state**.

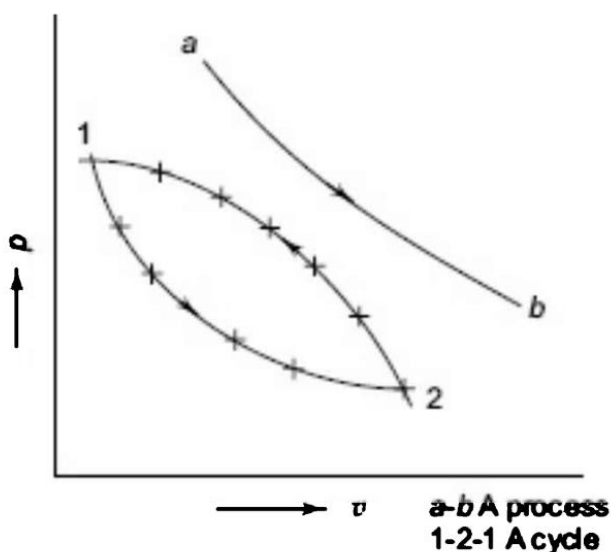


Figure 9 A process and a cycle

Any process or series of processes whose end states are identical is termed a **cycle**. Thus, a **thermodynamic cycle** is a series of processes (state changes) in which a working fluid begins at an initial state, moves through one or more intermediate states, and then returns to its initial state (Figure 9). If network is produced by the cycle, then it is labeled a power cycle. If work is added to the cycle and heat is pumped from a cold space to a hot space, then the cycle is a refrigeration cycle.

A closed system is said to undergo a cyclic process, or *cycle*, when it passes through a series of states in such a way that its final state is equal in all respects to its initial state

Thermodynamics properties

Every system has certain characteristics by which its physical condition can be described, e.g., volume, temperature, pressure etc. such characteristics are called properties. Thermodynamic properties are defined as characteristic features of a system capable of specifying the system's state. Some familiar properties are pressure P , temperature T , volume V , and mass m . The list can be extended to include less familiar ones such as viscosity, thermal conductivity, modulus of elasticity, thermal expansion coefficient, electric resistivity, and even velocity and elevation. The thermodynamic properties of the system are divided into two general classes: **Extensive** property and **Intensive** property.

Intensive properties

A quantity is called **intensive** when it is independent of the mass of a system, such as temperature, pressure, velocity, density, Specific Volume, Specific Entropy, Thermal conductivity, Thermal Expansion, Compressibility and many more. (Figure 10).

Extensive properties

Extensive properties are values that depend on the size (mass) or extent of a system such as mass, volume, momentum, energy, enthalpy, heat capacity, entropy, and total energy U . They are additive. An easy way to determine whether a property is intensive or extensive is to divide the system into two equal parts with an imaginary partition, as shown in Figure 10. Each part will have the same value of intensive properties as the original system, but half the value of the extensive properties.

Generally, uppercase letters are used to denote extensive properties (except mass m), and lowercase letters are used for intensive properties (except pressure P and temperature T).

Extensive properties per unit mass, volume, or the number of moles are called **specific properties**. In this case, we speak of a **specific quantity** or a **reduced extensive quantity**. Some examples of specific properties are specific volume ($v = V/m$) and specific total energy ($e = E/m$).

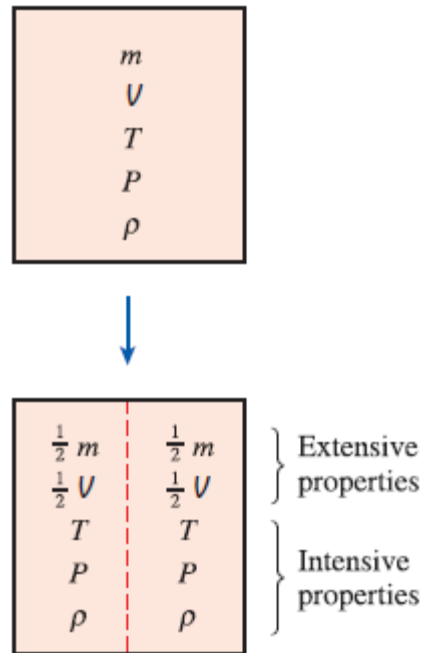


Figure 10 Criterion to differentiate intensive and extensive properties.

Definition of the Thermodynamics Properties

Temperature

Temperature (T), is the degree of hotness or coldness of the system. The absolute temperature of a body is defined relative to the temperature of ice; for SI units, the Kelvin scale. Another scale is the Celsius scale. Where The ice temperature under standard ambient pressure at sea level is: $0^{\circ}\text{C} \equiv 273.15 \text{ K}$ and the boiling point for water (steam) is: $100^{\circ}\text{C} \equiv 373.15 \text{ K}$. The imperial units of temperature is Fahrenheit where $T^{\circ}\text{F} = 1.8 \times T^{\circ}\text{C} + 32$

The concept of temperature was further refined by the introduction of the **zeroth law** of thermodynamics. It is called this because it was introduced about 100 years after the first and second laws.

Pressure

Pressure, P is defined as the normal force exerted per unit area of the surface within the system.

$$P = \frac{F}{A} \quad (1)$$

The unit of pressure in the SI system, is measured in Pascals (Pa). By definition, a pascal is a newton per square meter, or $1\text{Pa} = \frac{\text{N}}{\text{m}^2}$. Recall that the newton is a unit of force

The unit of pressure in the British system are pounds-force per square inch (lbf/in.²), also known as "psi."

Two other units, not part of the International System, continue to be widely used. These are the bar, where 1 bar = 10⁵ Pa = 0.1MPa and the standard atmosphere, where 1 atm = 101 325 Pa

For engineering work, pressures are often measured with respect to atmospheric pressure rather than with respect to absolute vacuum.

$$P_{abs} = P_{atm} = P_{gauge} \quad (2)$$

In fluids, gases and liquids, we speak of *pressure*; in solids this is *stress*. For a fluid at rest, the pressure at a given point is the same in all directions.

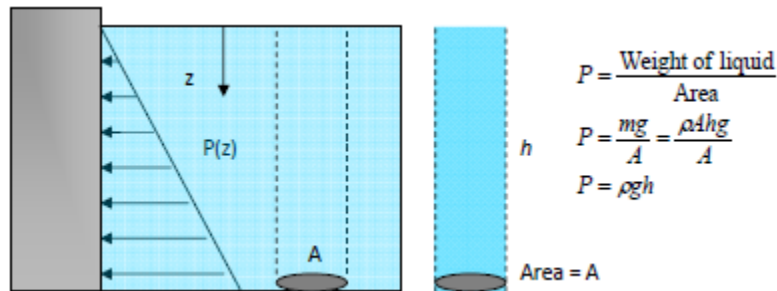


Figure 11 Pressure of a fluid at rest increases with depth (due to added weight), but constant in horizontal planes

Specific Volume (V) and Density (ρ)

The *specific volume* of a system is the volume occupied by the unit mass of the system. The symbol used is v and unit is, m³/kg. The symbol V will be used for volume. (Note that specific volume is *reciprocal of density*). Mathematically as in eq.1

$$v = \frac{\text{volume}}{\text{mass}} \quad (3)$$

Units are m³/kg

$$\text{It represents the inverse of the density, } v = \frac{1}{\rho} \quad (3b)$$

Internal energy (u)

Internal Energy (u) - The property of a system covering all forms of energy arising from the internal structure of the substance.

Enthalpy

Enthalpy (h) - A property of the system conveniently defined as $h = u + PV$ where u is the internal energy. SI Unit **kJ/kg**

Entropy

Entropy (s) - The microscopic disorder of the system. It is an extensive equilibrium property.

Specific Heats

The **specific heat** is defined as the energy required to raise the temperature of a unit mass of a substance by one degree. In general, this energy depends on how the process is executed. In thermodynamics, we are interested in two kinds of specific heats: **specific heat at constant volume C_V** and **specific heat at constant pressure C_P** .

Physically, the specific heat at constant volume C_V can be viewed as *the energy required to raise the temperature of the unit mass of a substance by one degree as the volume is maintained constant*. The energy required to do the same as the pressure is maintained constant is the specific heat at constant pressure C_P . specific heat at constant pressure C_P is always greater than C_V because at constant pressure the system is allowed to expand, and the energy for this expansion work must also be supplied to the system.

Quasi-static work, flow-work and specific heat

A system is in *thermodynamic equilibrium* if the temperature and pressure at all points are same. **Quasi-static (Quasi-equilibrium) process.** Quasi means ‘almost’. A quasi-static process is also called a *reversible process*. It can be viewed as a sufficiently slow process that allows the system to adjust itself internally and remains infinitesimally close to an equilibrium state at all times. Quasi-equilibrium process is an idealized process and is not a true representation of the actual process. We model actual processes with quasi-equilibrium ones. Moreover, they serve as standards to which actual processes can be compared.

To consider how a gas (or liquid) might be expanded or compressed in a quasiequilibrium fashion, refer to Figure.12 Illustration of a quasiequilibrium expansion or compression, which shows a system consisting of a gas initially at an equilibrium state. As shown in the figure, the gas pressure is maintained uniform throughout by a number of small masses resting on the freely moving piston.

Imagine that one of the masses is removed, allowing the piston to move upward as the gas expands slightly. During such an expansion the state of the gas would depart only slightly from equilibrium. The system would eventually come to a new equilibrium state, where the pressure and all other intensive properties would again be uniform in value. Moreover, were the mass replaced, the gas would be restored to its initial state, while again the departure from equilibrium would be slight. If several of the masses were removed one after another, the gas would pass through a sequence of equilibrium states without ever being far from equilibrium. In the limit as the increments of mass are made vanishingly small, the gas would undergo a quasiequilibrium expansion process. A quasiequilibrium compression can be visualized with similar considerations.

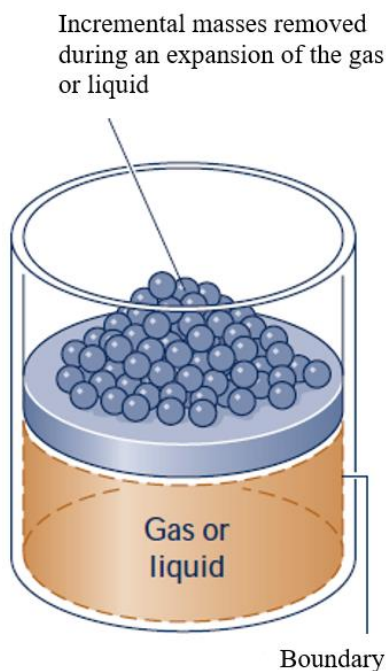


Figure.12 Illustration of a quasiequilibrium expansion or compression

Laws of thermodynamics

Zeroth law of thermodynamics

If an object with a higher temperature comes in contact with a lower temperature object, it will transfer heat to the lower temperature object. The objects will approach the same temperature, and in the absence of loss to other objects, they will maintain a single constant temperature. Therefore, thermal equilibrium is attained.

The “zeroth law” states that if two systems are at the same time in thermal equilibrium with a third system, they are in thermal equilibrium with each other.

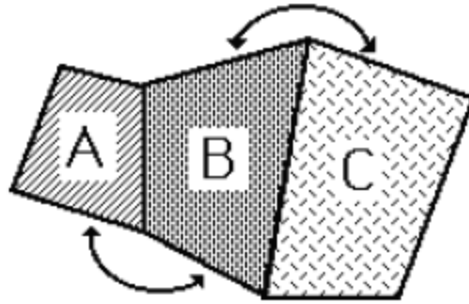


Figure13 Analogy of the Zeroth Law of Thermodynamics

If A and C are in thermal equilibrium with B, then A is in thermal equilibrium with B. Practically this means that all three are at the same temperature, and it forms the basis for comparison of temperatures.

The Zeroth Law states that:- “*two systems which are equal in temperature to a third system are equal in temperature to each other*”. This law deals with thermal equilibrium and establishes a concept of temperature.

First law of thermodynamics

The first law of thermodynamics is the application of the conservation of energy principle. States that *energy can be neither created nor destroyed during a process; it can only change forms*. This simply means that during an interaction, energy can change from one form to another but the total amount of energy remains constant. This law throws light on concept of internal energy.

Example of first law applicable to a closed system. Consider a closed system where there is no flow into or out of the system, and the fluid mass remains constant. For such system, the first law statement is known as the Non-Flow Energy Equation, or NFEE abbreviated, it can be summarised as follows:

$$\Delta U = Q - W \quad 4$$

The first law makes use of the key concepts of internal energy (ΔU), heat (Q), and system work (W).

Second law of thermodynamics

Second law of thermodynamics states that *heat will flow naturally from one energy reservoir to another at a lower temperature, but not in opposite direction without assistance*. This is very important because a heat engine operates between two energy reservoirs at different temperatures. Further the first law of thermodynamics establishes equivalence between the quantity of heat used

and the mechanical work but does not specify the conditions under which conversion of heat into work is possible, neither the direction in which heat transfer can take place. This gap has been bridged by the second law of thermodynamics.

Third law of thermodynamics

The third law of thermodynamics states “When a system is at zero absolute temperature, the entropy of system is zero”.

All the laws of thermodynamics are based on experimental observations and have no mathematical proof. Like all physical laws, these laws are based on logical reasoning.

Steady flow energy equation without proof

Assignment 1

Briefly answer the following questions:

1. What is thermodynamics? List 6 the application areas of thermodynamics.
2. Define a thermodynamic system. Differentiate between open system, closed system and an isolated system with examples. Support your explanation with sketches.
3. Briefly explain the term working fluid in thermodynamics with examples
4. Differentiate between intensive and extensive properties of thermodynamics with 5 examples each with their SI units. E,g pressure pascal (N/m^2)
5. Explain the following terms: (i) State, (ii) Process, (iii) Cycle and (iv) specific heat at constant volume and pressure.
6. State the four laws of thermodynamics.
7. What is a quasi-static process?

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